

Introduction

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The subject data structure deals with the study of methods, techniques, and tools to organize or structure data in computer memory.

Now before defining data structure we should know what is data and what is the difference between data and information.

Data and Information

Data is a plural of datum, which is originally a Latin noun meaning something given.

The Oxford Dictionary meaning of data is:

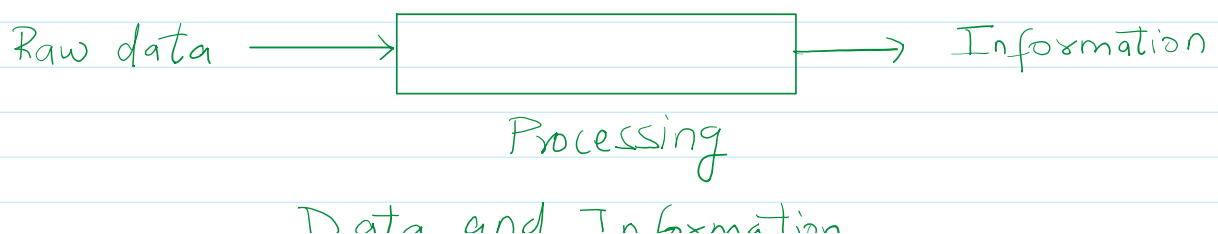
- i. Facts or statistics used for reference or analysis.
 - ii. The quantities, characters, or symbols operated by a computer.
- For our purpose, the second meaning is more important. Therefore we can say that:

The data represents quantities, characters or symbols on which operations are performed by a computer, stored and recorded on either magnetic, optical or mechanical recording media, and transmitted in the form of digital electrical signals.

Definition: Data is the basic entity or fact that is used in calculation or manipulation process. Data is commonly processed by some stages. Unprocessed data or raw data is a collection of numbers, characters that may be considered as an input of a stage, and processed data is the output of the stage.

There are two types of data such as numerical and alphanumerical data. Data may be single value or a set of values and it is organized in a particular fashion. This organization or structuring of data will have a profound impact on the efficiency of the program.

Most of the individual consider that the term "data" and "information" are interchangeable and meaning the same thing. However, there is a distinct difference between the two words. Data are raw facts without context, whereas information is data with context. Data are an abstraction of information in problem solving system. Data requires interpretation to become an information.



Data and Information

Representation of Data

Almost all the high-level languages examples Fortran, Cobol, C, Pascal, C++, Java, etc. Allow data to be represented in computer memory of mainly two types:

- i. Numeric data.
- ii. Alphanumeric data.

Most of the high. Level languages classified numerical data into 2 types:

- i. Integer
- ii. Floating point

You can see that in C language the following data types are used to represent numerical data.

The ranges of numerical data types

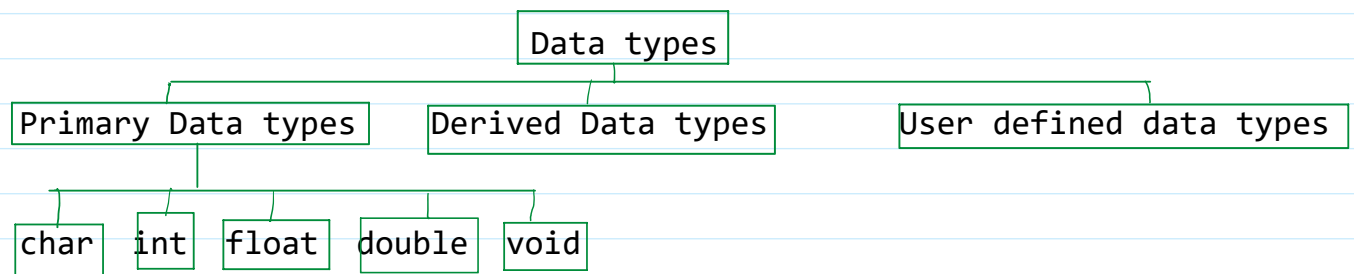
Data types	Memory requirement	Ranges
char or signed char	1 byte	-128 to 127
unsigned char	1 byte	0 to 255
short signed int or short int or signed int or int	2 bytes	-32768 to 32767
long signed int or long int	4 bytes	-2147483648 to 2147483647
signed int or int(for 32 bit compiler)	4 bytes	-C to 2147483647
unsigned int	2 bytes	0 to 65535
long unsigned int	4 bytes	0 to 4294967295
float	4 bytes	-3.4e38 to 3.4e38
double	8 bytes	-1.7e308 to 1.7e308
long double	10 bytes	-1.7e4932 to 1.7e4932

Data Type

Generally, computer programs do exist for a single purpose: how to process data. The type of data, the format of data that is going to be returned and the correctness of the processing are primary concerns of a computer program. When a program is written, how the computer handles the data internally is usually a secondary concern.

Definition: A data type refers to the type of data that variables hold.

Now, depending on the representation of different forms of data, different data types are used. The data types are name given to a set of variables which have common properties. A data type refers to the type of data that variables hold.



Data types

The C language supports three classes of data types:

- i. Primary or Basic data types
- ii. Derived data types
- iii. User-defined data types

Basic or primary data types are language built - in. The C language supports five primary or basic data types. They are:

- i. char
- ii. int
- iii. float
- iv. double
- v. void

Data Structure

Data structure is the representation of the logical relationship between individual elements of data. In computer science, data structure is defined as a mathematical or logical model of organizing the data items into computer memory in such a way that they can be used efficiently. The purpose of studying data structure is to learn how to organize data into memory so that it can be accessed quickly and conveniently.

Data structure refers to study of data and representation of data objects within the program, that is the implementing of structured relationship among different data objects.

Examples: list, stack, queues, heaps, search trees, hash tables etc.

Different set of operations can be performed on different data structures. The operations that can be performed on different data elements within a data structures are accessing, traversing, inserting, deleting, modifying, etc. While writing a program, a minimal data

structure must be chosen that supports all the operations as per need.

Data structure can be used for the following purpose:

- i. Organizing the data - How data items are organized in the main memory?
- ii. Accessing methods - How data items can be accessed?
- iii. Specifying the degree of associativity - How data items are interrelated?
- iv. Processing alternatives for data - How many different ways are there in which these data items can be processed?

A program is a set of instructions which involve a computer performing some kind of computation or algorithm. Data structure affects the design of both structural and functional aspects of a program. To implement a program of an algorithm, we should select an appropriate data structure for that algorithm. Therefore, the programs inherited by an algorithm and its associated data structure.

Algorithm + Data Structure = Program

Programming languages provide facilities for representatives of algorithms and data. High level programming languages like C facilitates structured and modular programming by providing algorithm structures.

Classification of data structure

Data structures can be classified into two categories:

- i. Primitive data structure.
- ii. Non - primitive data structure.

Primitive data structure

The primitive data structures are defined that can be manipulated or operated by the machine instruction. There are numerous types of data structures, generally built upon simpler primitive data types called primitive data structures, which are represented in computer memory.

Example: Integer, floating point, characters, pointer, boolean, etc. are some of the different primitive data structures.

In C language, the different primitive data structures are defined using the data type such as int, care, float, double, etc.

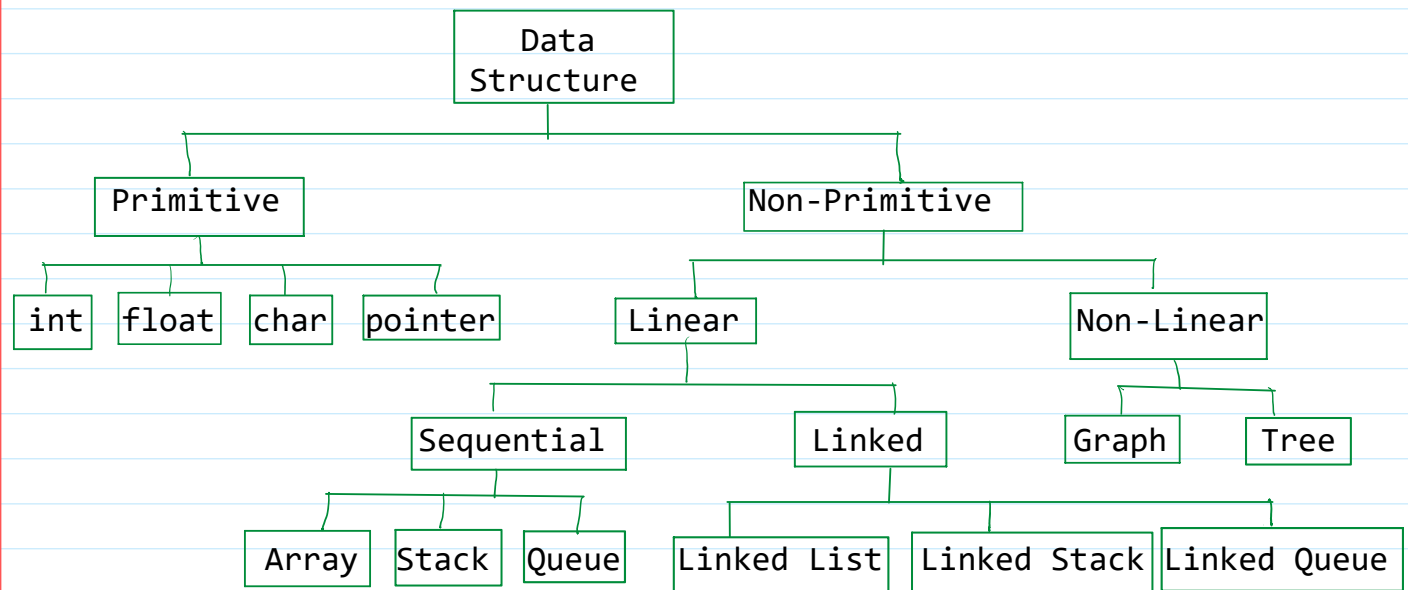
Non primitive data structure

The non primitive data structures are a data structure that cannot be manipulated or operated directly by the machine instructions. These are more sophisticated data structures. Non - primitive data structures are derived from the primitive data structures. These are a group of homogeneous or heterogeneous data items.

Example: Arrays, structure, stack, queues, linked list, etc.

The non primitive data structures are classified into two categories:

- i. Linear data structure.
- ii. Non-linear data structure.



Classification of Data Structures

Linear Data Structures

The data structure is linear if every data item is related to the previous and the next items. Example array linked list.

In the linear data structures, data items are arranged in memory in a linear sequence and data items are accessed linearly. The traversing of the linear data structure is exactly once.

Linear data structures are of two types:

- a. Sequential
- b. Linked

Sequential data structures are based on arrays, where objects are stored in a sequence of consecutive memory locations. Example: Arrays, Stacks, Queues

Linked data structure is a data structure which consists of a set of nodes linked together and organized with links.

Example: linked list, linked stack, linked queues

Non-linear Data Structures

A data structure is nonlinear if every data item attached to many other data items in a specific way to reflect relationship. (example tree) In nonlinear

data structure, the data items are not in sequence. That is, insertion and deletion are not possible in a linear manner. The traversing of a non linear data structure is always more than one.

Example: Graph, Trees

Static Data structure

The starting data structure. Is a kind of data structure in which one's memory space is allocated. It cannot extend. That is, the memory allocation for the data structure takes place at compile - time that cannot be changed afterwards.

Example: Array

Dynamic Data Structure

Dynamic data structure is another kind of a data structure which can be extended or shrink during the execution. That is the memory allocation as well as memory deallocation for the data structure takes place at runtime and allocates memory as required amount at any time.

Example: linked list, stack, queue, tree

Application of data structure

Different data structure is used in real life, such as the representation of an image in the form of a bitmap, implement printer spooler so that jobs can be printed in the order of their arrival, store information about directories and file system etc. Data structure is used in various fields of computer science, such as:

- Compiler design.
- Operating system.
- Database management system.
- Statistical analysis package.
- Numerical analysis.
- Graphics.
- Artificial intelligence.
- Simulation.

Different kinds of data structures are suitable for different kinds of application. Some data structures are highly specialized for specific task. For example database use B tree indexes for small percentage of data retrieval and compilers and database use dynamic hash tables to look up tables. Operating system uses queues for process management, I/O request handling.

Abstract Data type:

But data type refers to the type of data that variables hold. Thus integer real characters are referred to as a primitive data types.

Data Object Represents an object having a data. The study of classes of objects whose logical behavior is defined by a set of values and a set of operations.

Definition: An abstract data type (ADT) describes the data object which constitute the data structure and the fundamental operations supported on them.

An ADT promotes data abstraction and focuses on what a data structure does rather than how it implements (does)

- An ADT is a conceptual model of information structure.
- An ADT specifies components, their structuring relationship and a list of operations that are allowed to perform.
- It is just a specification, no design or implementation information is included.
- Specification involves the what, the operation, not the how.
- ADT's are generalization of primitive data types.

A data structure is a design representation of an ADT.

- The same ADT may be represented by several data structures.
- There are many data structures correspond to the ADT set.

Operations perform on data structures

Data are processed by means of certain operations which appear in the data structure. The particular data structure is chosen largely depends on the frequency of the operation that needs to be performed on the data structure. Different kinds of operations are to be performed on data structures.

Operations on data structures

Operation	Description
Creation	Allocation of memory for the data structure. The creation of data structure may takes place either during compile time or during runtime.
Insertion	Insert a data item in the data structure.
Deletion	Delete a data item from the data structure.
traversing	Accessing and processing each data item for the data structure exactly once.
Searching	Find the location of the key value within the data structure.
Sorting	Arranging all the data items in a data structure either in ascending or in descending order or in lexicographical order (for strings).
Merging	Combining the data items of two different sorted list into a single sorted list.

Overview of different data structures

Array: An array is a collection of same type of data items which are stored in consecutive memory locations under a common name. In Arrays, there is always a

fixed relationship between the addresses of two consecutive elements, as all the items of an array must be stored contiguously.

Stack: A stack is a collection of elements into which new elements may be inserted and from which elements may be deleted only at 1 end called the top of the stack. Since all the insertion and deletion in a stack is done from the top of the stack, the last added element will be the first to be removed from the stack. That is the reason why stack is also called last in, first out (LIFO) data structure.

Queue: A queue is a homogeneous collection of elements in which deletions can take place only at the front end, known as dequeue and insertion can take place only at rear and known as enqueue. The element which inserts in the queue first will delete the queue first. In this order, a queue is called first - in - 1st - out (FIFO) system.

Linked List: The linked list is an ordered collection of finite homogeneous data elements called node, where the linear order is maintained by means of links or pointers. In linked list, data items may be scattered arbitrarily all over the memory. In a linked list, there is no relationship between the address of elements. Each element of a linked list must store explicitly the address of the element next to it.

Tree: The tree is a non - linear data structure. A tree may be defined as non - empty finite sets of nodes such that:

- i. There is a specially designed node is called a root node.
- ii. The remaining nodes are partitioned into zero or more disjoint trees. $T_1 T_2 \dots T_N$. Are called subtrees of the root R.

Graph: The graph is another non - linear data structure. A graph G is defined as an ordered set G equal to (VE), consists of finite non - empty set of objects. V where $V(G) \{V_1, V_2, V_3 \dots V_N\}$ Are called vertices. (or nodes or points) and another set E where $E(G) = \{e_1, e_2, e_3, \dots e_m\}$ whose elements are called edges, that connects these vertices.